Diabetic Retinopathy Screening

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Diabetic Retinopathy Screening

Disease

Diabetic retinopathy (DR) is the leading cause of preventable blindness.[1] Of working-age Americans diagnosed with Diabetes Mellitus, it is estimated that about 1/3 have diabetic retinopathy (DR) and 4.4% have vision threatening retinopathy.[2] [3] For more information on diabetic retinopathy disease, grading, and pathophysiology, see Diabetic Retinopathy and Diabetic Retinopathy Pathophysiology.

Risk Factors

Independent risk factors for DR included diabetes duration, hemoglobin A1c, serum glucose, systolic blood pressure, and Indian ethnicity based on the Singapore Eye Disease Study.[4] Duration of diabetes is a major risk factor, and is the main criteria utilized to decide when to begin DR screening. After 5 years, approximately 25% of type 1 patients will have retinopathy. After 10 years, almost 60% will have retinopathy, and after 15 years, 80% will have retinopathy.[5] Proliferative diabetic retinopathy was present in approximately 50% of type 1 patients who had the disease for 20 years. [6] In the Los Angeles Latino Eye Study (LALES) and in Proyecto VER (Vision, Evaluation and Research), 18% of participants with diabetes of more than 15 years' duration had PDR, and there was no difference in the percentage with PDR between those with type 1 and type 2 diabetes.[7][8]

Screening Methods

The most recent guidelines for DR screening were released by the International Council of Ophthalmology (ICO) and American Diabetes Association (ADA) in 2018.[9] This article focuses on DR screening in the United States. Other guidelines may vary by country and availability of resources.[10] [11] [12]

The method used to screen for DR is dependent on resource settings. The 2018 ICO/ADA guidelines state that adequate DR screening should include a visual acuity exam and a retinal examination.

Adequate visual acuity screening includes at least one of the following:

(1) refracted visual acuity examination using 3- or 4-m visual acuity lane and a high-contrast visual acuity chart

(2) presenting visual acuity examination using a near or distance eye chart and pin-hole option if visual acuity is reduced

(3) presenting visual acuity examination using a 6/12 (20/40) equivalent handheld chart consisting of at least 5 standard letters or symbols and a pin-hole option if visual acuity is reduced.

Adequate retinal examination includes at least one of the following:

(1) direct or indirect ophthalmoscopy or slit-lamp biomicroscopic examination of the retina

(2) retinal (fundus) photography, including any of the following: 30° to wide field, monophotography or stereophotography, and dilated or undilated photography.[9]

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The retinal examination can be done with or without optic coherence tomography (OCT). Retinal examination may not need to be performed by individuals with a medical degree, so long as they are trained to perform ophthalmoscopy or retinal photography and can assess disease severity.[9]

Screening Rates in the United States and Impact of Social Determinates of Health

Despite the high prevalence of DR, only 62.3 percent of patients with diabetes in the United States receive annual screening exams.[13] [14] Screening rates within subsets of the US population vary greatly. Social determinates of health have an important role in DR screening rates. Numerous studies have identified lower educational status, lower income, minority race, recent immigration, residence in a rural community, and lack of health insurance with significantly decreased rates of DR screening.[2][15] [16] [17] [18] [19] [20] [21] [22] [23] Income is a particularly important factor, as patients of lower socioeconomic status are less likely to receive eye care.[24] Patients of minority race and ethnicity, including Black and Hispanic patients, are less likely to be aware of their diagnosis of DR and to receive screening.[25] [26] Residence in a disadvantaged neighborhood is associated with decreased likelihood of adherence to DR screening.[27] Health insurance status was found to have the most profound effect on DR screening rates, with 76% of patients with insurance receiving proper screening while only 36% of uninsured patients received proper screening.[16]Patients who do not receive eye care often lack trust in the medical system.[28]

Increasing DR screening rates is an important goal of the US Department of Health and Human Services Healthy People 2030 campaign. The DR goals for Healthy People 2030 include increasing DR screening in people with diabetes from 62.3 percent to 67.7 percent and reducing vision loss from DR in people with diabetes from 33 percent to 16.5 percent.[14][29]

Initial Screening and Referral

Based on recent guidelines by the AAO Diabetic Retinopathy Preferred Practice Pattern for the recommended time of first retinal examination for patients with diabetes: (Flaxel et al. 2020, Diabetic Retinopathy Preferred Practice Pattern[30])

- Type 1 DM: 1st Retinal exam 3-5 years after diagnosis
- Type 2 DM: 1st Retinal exam at the time of diagnosis
- Pregnancy (with type 1 or type 2): 1st Retinal exam soon after conception and early in the 1st trimester of pregnancy

An eye examination is not required if patients without DM newly develop gestational diabetes during pregnancy.

Screening follow-up depends on the severity of disease as well as the available referral resources in a given geographic location. Table 1 summarizes the screen and referral guidelines for DR according to the ICO/ADA 2018 guidelines in high resource settings. DM without DR and mild NPDR does not need to be referred to an ophthalmologist. In addition, high-resource settings may employ the use of optical coherence tomography (OCT) to further satisfy patients that have diabetic macular edema to further determine follow-up intervals.

Table 1: ICO/ADA 2018 DR Screening and Referral Guidelines for High Resource Settings

Classification	Re-examination or Next Screening Schedule	Referral to Ophthalmologist
Diabetic Retinopathy (DR)		
No apparent DR, mild nonproliferative DR, and no DME	Re-examination in 1–2 yrs	Referral not required
Mild nonproliferative DR	6–12 mos	Referral not required
Moderate nonproliferative DR	3–6 mos	Referral required
Severe nonproliferative DR	<3 mos	Referral required
Proliferative DR	<1 mo	Referral required
Diabetic Macular Edema (DME)		
Non–center-involving DME	3 mos	Referral required
Center-involving DME	1 mo	Referral required

Resource-Limited Settings and Cost Effectiveness

Dilated fundus exam performed by an ophthalmologist is considered the gold standard method for diagnosing DR and monitoring patients at risk of developing DR; however, a yearly exam in a resourced-limited setting would prove unfeasible given the growing gap in access to eye care professionals.[31] With recent advances in technology, most screening programs are transitioning to retinal photography-based screening.[32] Recent studies have indicated that photography screening programs are cost effective for populations with over 3,500 individuals aged 50-80 and can generate a savings of $127 per individual with diabetes screened over their lifetime.[33] Kahn et al. demonstrated this cost savings could be achieved with only 65% of individuals completing screening.[34]

Overall, the ICO recommended screening and referral guidelines only differ to a minor extent for limited resource areas. Longer screening intervals can be used for low-risk individuals in the mild and moderate NPDR categories, while more severe disease has the same recommended interval (Table 2).[9] Additionally, non-center involving DME treatment may be delayed in settings where access to laser therapy is limited.

Table 2: ICO/ADA 2018 DR Screening Follow-up Guidelines for Low-Intermediate Resource Settings

Classification	Re-examination or Next Screening Schedule	Referral to Ophthalmologist
Diabetic Retinopathy (DR)		
No apparent DR, mild nonproliferative DR, and no DME	Re-examination in 1–2 yrs	Referral not required
Mild nonproliferative DR	1—2 yrs	Referral not required
Moderate nonproliferative DR	6-12 mos	Referral required
Severe nonproliferative DR	<3 mos	Referral required
Proliferative DR	<1 mo	Referral required
Diabetic Macular Edema (DME)		
Non–center-involving DME	3 mos	Referral not required (referral recommended if laser sources available)
Center-involving DME	1 mo	Referral required

Retinal Photography and Tele-retina

Tele-retina has been proposed as a cost-effective alternative to examination by an ophthalmologist[33][35] whereby retinal images taken at one site are

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Non-mydriatic photography can be accomplished with minimal training in the primary care setting[34] and may demonstrate similar sensitivity and specificity (78-98%, 86-90%, respectively) to a dilated fundus exam performed by an ophthalmologist (84-92%, 92-98%, respectively).[36]

Digital retinal imaging takes approximately 15 to 20 minutes. The images can be analyzed remotely by a grader to identify the presence of referable or vision-threatening disease. When compared with dilated ophthalmoscopic examination or the gold standard seven-field stereoscopic fundus photography for retinopathy screening, nonmydriatic digital photographs have historically demonstrated good sensitivity and specificity for detecting DR[37][38] [39].

Ultra-wide field imaging

Because retinal features in diabetic retinopathy can often be located in the periphery of the retina, improved detection and screening methods may be achieved with the use of ultra-wide field imaging.

There are currently three methods for obtaining peripheral views of the retina: 1) creating montages from individually acquired conventional images with steering, 2) using a special high-index lens (contact or noncontact) in conjunction with a conventional camera, and 3) using a dedicated wide-angle camera system. Ultra-wide field (UWF) retinal photography has developed from confocal scanning laser ophthalmoscopy to provide peripheral views of nearly the entire retina (up to 200 degrees, representing approximately 82.5% of the total retina surface) in a single capture.

Most studies comparing ultra-wide field photography to traditional methods of detecting DR have determined the sensitivity and specificity be similar. Friberg et al. compared nonmydriatic UWF images to the clinical exam and found that UWF detected DR as a general diagnosis with 94% sensitivity.[40] Neubauer et al. found that for cases more severe than mild NPDR, UWF had a sensitivity of 94% and specificity of 100% when compared to onsite mydriatic ophthalmologic examination[41]. In a study of 206 eyes with DR in 2013, Silva et al. showed that UWF photography matched the standard EDTRS grade of severity in 80% of eyes, and was within one level in 94.5% of eyes[42]. In fact, UWF may reveal additional retinal abnormalities not originally included in the ETDRS grading criteria for diabetic retinopathy[43]. An added benefit of UWF is that image acquisition time is typically less than half that of mydriatic ETDRS photography.[44]

It is possible that UWF photography may allow for the detection of new biomarkers for DR. Dodo et al. showed that ultra-wide field imaging could also detect early NPDR by visualizing so-called “white dots” on the retina, which were used as markers for significantly nonperfused areas.[45] They found that the number of white dots visualized on UWF imaging generally increased with disease severity. there is evidence that, even without angiography, this ultra-wide field imaging modality may serve as a useful screening tool and allow for more accurate grading of diabetic retinopathy.

Artificial Intelligence

While there are benefits including reduced need for dilating medications and easy image acquisition with retinal photography, there are reports of high technical failure rates and the continued reliance on a trained image grader.[33] Consistent follow up on screening results is necessary for an effective tele-retina program, and health systems must be prepared for the upfront costs of implementing these systems.[46] However, the wider implementation of retinal photography in various settings such as primary care clinics, remote areas without ophthalmologists, and underserved populations is sure to enhance DR detection and improve access to care. Standardizing a consistent method to analyze retinal photographs are of great interest, and early artificial intelligence (AI) methods seem to hold promise. These technological solutions reduce the need for trained graders in the primary care setting and enable more efficient screening. AI reduces burden of manual review of fundus photography, using technology to automatically screen fundus images for presence of disease. A systemic review of AI performance in DR screening demonstrates these technologies have high sensitivity (87.0%-95.2%) but generally a low specificity (49.6–68.8%).[47] Predictive modeling is increasingly being used to identify patients at risk of DR progression and define personalized screening intervals.[48] [49] While the personalized screening intervals may perform better than the low-resource extended intervals, they have a slightly higher implementation cost.[49] Patients’ acceptability, patients’ confidentiality, and medico-legal challenges are issues facing widespread adoption of these technologies.[50] Nevertheless, the need for more efficient screening for DR in a growing population with Diabetes Mellitus shows promise for such technologies.

The FDA in 2018 permitted marketing of the first authorized AI device to detect DR, Idx-DR. This software utilizes retinal images from Topcon NW400, a nonmydriatic, non-ultrawide field camera. Two 45-degree photographs, disc-centered and macula-centered, are utilized for analysis. If the images are of sufficient quality, the software provides the doctor with one of two results: (1) “more than mild diabetic retinopathy detected: refer to an eye care professional” or (2) “negative for more than mild diabetic retinopathy; rescreen in 12 months.” This automated evaluation is limited in pregnant women, in whom retinopathy can progress rapidly and patients with previously diagnosed severe NPDR, PDR, DME, or other significant ocular history[51]. The Iowa Detection Programme (IDP) studied earlier versions of IDx-DR and have shown comparable detection of hemorrhages, exudates, cotton wool spots, neovascularization, and irregular lesions similar to human graders. [52] Subsequently, IDx-DR improved upon previous algorithms with the inclusion of deep learning features, with recent studies citing a sensitivity of 96.8% and improved specificity of 87% for this system to detect referrable diabetic retinopathy[53]. The published results of the first preregistered clinical trial of an AI system by Abramoff et al. led to this landmark FDA approval of the first fully-autonomous AI diagnostic system. [54]

Several alternative systems are in development currently, and future algorithms may employ the use of UWF cameras to include peripheral retinal analysis[55]. The implementation of camera-phone attachments may also allow widespread screening of patients even in their own home, especially in resource-poor settings[56]. Patients’ acceptability, patients’ confidentiality, and medico-legal challenges are issues facing widespread adoption of these technologies.[50] Nevertheless, the need for more efficient screening for DR in a growing population with Diabetes Mellitus shows promise for these rapidly evolving technologies.

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